PS Analysis in Data Science: Problem Set 14

Problem 1. Consider $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = (x + 1)^2 + 3y^2$$

(a) Determine all local extrema of f and check by the Hessian if it is a local minimum or local maximum.

(b) Consider the constraint problem $g(x) = 0$, for which $g : \mathbb{R}^2 \to \mathbb{R}$ given by

$$g(x, y) = x^2 + y^2 - 1$$

Compute all local constraint extrema.

(c) Check by the Hessian of the Lagrangian if each local constraint extremum is a local constraint maximum or minimum.

(d) Visualize f and g is Julia, so that you "see" the local (constraint) extrema.

(e) Paramterize the circle by $\gamma : [0, 1] \to \mathbb{R}^2$,

$$\gamma(t) = \begin{pmatrix} \cos(2\pi t) \\ \sin(2\pi t) \end{pmatrix}$$

and plot the function $t \mapsto f(\gamma(t))$. Does this plot confirm the local constraint extrema?

Problem 2. Let X and Y be metric spaces. Verify the following claim: a function $f : X \to Y$ is continuous if and only if, for every closed subset $U \subseteq Y$, the pre-image $f^{-1}(U)$ is closed.

Problem 3. Let $\mathcal{C}([0, 1]) := \{f : [0, 1] \to \mathbb{R} \mid f \text{ continuous}\}$ with the norm $\|f\|_\infty = \sup_{x\in[0,1]} |f(x)|$. Define the map

$$\Phi : \mathcal{C}([0, 1]) \to \mathcal{C}([0, 1]), \qquad (\Phi(f))(t) = \frac{t + f(t)^2}{4}, \quad t \in [0, 1].$$

(a) Show that $\mathcal{C}([0, 1])$ is a Banach space.

(b) Determine the Fréchet derivative $\Phi'(f)$.

(c) Show that

$$\|\Phi'(f)\|_{op} \leq \frac{1}{2}, \qquad \forall\, f \in \mathcal{C}([0, 1]), \quad \|f\|_\infty \leq 1.$$

(d) Prove that Φ has a fixed point and determine it.